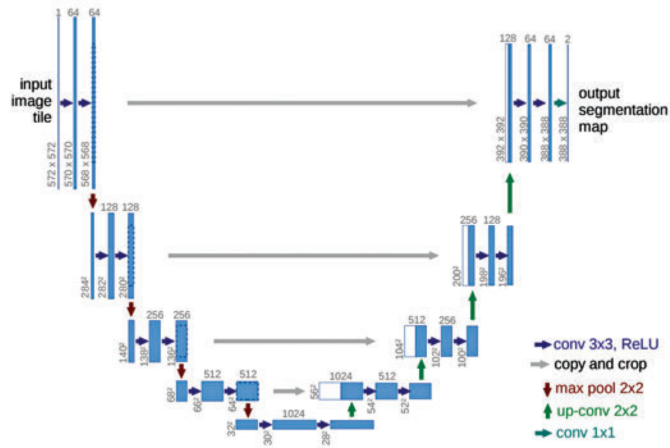

Fig. 1. U-net architecture (example for 32x32 pixels in the lowest resolution). Each blue box corresponds to a multi-channel feature map. The number of channels is denoted on top of the box. The x-y-size is provided at the lower left edge of the box. White boxes represent copied feature maps. The arrows denote the different operations.

the downsampling ResNets of the encoder and the upsampling ResNets of the decoder. Furthermore, the Stable Diffusion U-Net can condition its output on text embeddings through cross-attention layers, which are usually added between ResNet blocks in both the encoder and decoder parts of the U-Net.

What is the text encoder?

The text-encoder converts the input prompt, such as “An astronaut riding a horse,” into a latent embedding space that can be processed by the U-Net. Typically, it is a transformer-based encoder that maps a sequence of input tokens to a sequence of latent text-embeddings.

Stable Diffusion takes inspiration from Imagen and does not train the text-encoder during training. Instead, it utilizes the pre-trained text encoder CLIPTextModel from CLIP.

Latent diffusion is fast and efficient because it operates on a lower dimensional space, resulting in reduced memory and compute requirements compared to pixel-space diffusion models. In Stable Diffusion, the autoencoder used has a reduction factor of 8, which means that an image of size (3, 512, 512)